

Prototype embeddings for immune receptors: geometry, statistics, and the TCREMP coordinate system

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Appendix T — TCREMP embedding geometry

Abstract

TCREMP [1] embeds an immune receptor as the vector of its alignment distances to a fixed set of *prototype* receptors sampled from the V(D)J recombination model [2, 3]. This appendix gives the mathematics behind that construction. We first develop the general theory of *prototype* (landmark) embeddings of a metric space — their Lipschitz and isometry properties, the negative-type condition under which the alignment dissimilarity is an honest squared Hilbert distance [4], the energy functional that makes embedding-space Euclidean distance track pairwise alignment distance, and a landmark sample-complexity bound (§T.1–§T.2). We then specialise to immune receptors: taking the prototype measure to be the generation distribution P_{gen} turns each coordinate into a Monte-Carlo sketch of a receptor’s relationship to the generative repertoire, which explains why the choice of prototype source scarcely matters and yields the observed Gamma / extreme-value distribution laws (§T.3). The three-component structure of the TCR embedding and paired-chain concatenation are read off as a product metric, with the germline blocks provably low-rank (§T.4); somatic hypermutation is a bounded perturbation that explains why IGH is the hardest chain (§T.5); and the pushforward of P_{gen} onto embedding space is the null model for density-based background subtraction (§T.6). Every quantitative claim is reproduced in-repo (§T.8, Table 1) by `mir/bench/theory.py`.

T.1. FOUNDATIONS OF PROTOTYPE EMBEDDINGS

SETUP. Let (X, d) be a (semi)metric space of receptor sequences: d is symmetric, $d(x, x) = 0$, and — as we verify for the alignment dissimilarity in §T.2 — non-negative. Fix an ordered set of *prototypes* $P = \{p_1, \dots, p_K\} \subset X$. The **prototype embedding** is the landmark map

$$\varphi : X \rightarrow \mathbb{R}^K, \quad \varphi(x) = (d(x, p_1), d(x, p_2), \dots, d(x, p_K)), \quad (1)$$

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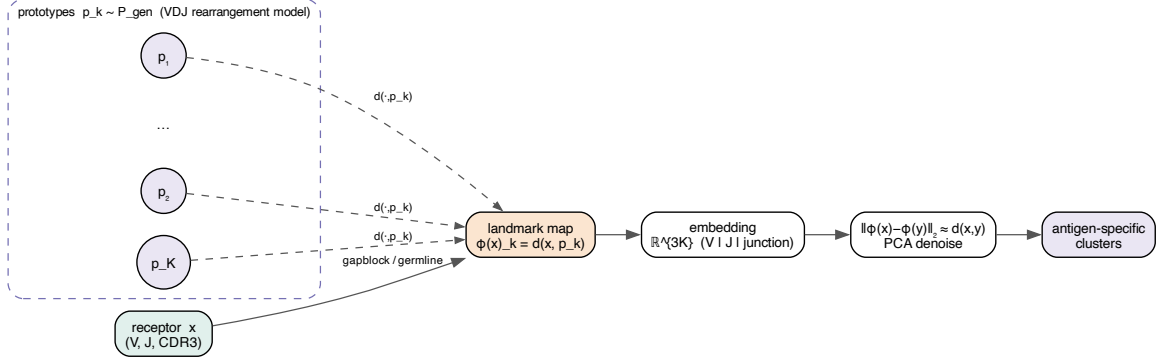


Figure 1: The TCREMP prototype-embedding pipeline. A receptor x is mapped to the vector of its alignment distances to prototypes p_k sampled from the recombination model P_{gen} ; Euclidean distance in the resulting $\mathbb{R}^{3K}$ coordinate system approximates pairwise alignment distance (Prop. 1–3), and specificity groups become compact clusters.

and the induced **embedding distance** is $D(x, y) = \|\varphi(x) - \varphi(y)\|_2$. Figure 1 shows the pipeline. In TCREMP the prototypes are drawn from P_{gen} , d is a BLOSUM alignment dissimilarity, and φ carries three coordinate blocks per prototype (§T.4); the foundational results below hold for any (X, d, P) .

NON-EXPANSIVENESS. The single structural fact from which everything else follows is that each coordinate of φ is 1-Lipschitz.

PROPOSITION 1 (Lipschitz coordinates and the upper bound). *For every prototype p and all $x, y \in X$, $|d(x, p) - d(y, p)| \leq d(x, y)$. Consequently*

$$\|\varphi(x) - \varphi(y)\|_\infty \leq d(x, y) \quad \text{and} \quad D(x, y) = \|\varphi(x) - \varphi(y)\|_2 \leq \sqrt{K} d(x, y). \quad (2)$$

Proof. The triangle inequality gives $d(x, p) \leq d(x, y) + d(y, p)$ and $d(y, p) \leq d(x, y) + d(x, p)$, so $|d(x, p) - d(y, p)| \leq d(x, y)$. Taking the maximum over $p \in P$ gives the ℓ^∞ bound; summing the squares of K terms each $\leq d(x, y)^2$ and taking the root gives the ℓ^2 bound. $\square$ $\square$

Proposition 1 is the rigorous half of the headline “embedding distance $\leq$ alignment distance” claim: no pair is ever *farther* apart in embedding space (up to $\sqrt{K}$) than its alignment distance. The reverse — that close-in-embedding implies close-in-alignment — is not automatic and is where the prototypes must do work.

EXACTNESS IN THE LIMIT. With enough prototypes the ℓ^∞ bound is tight.

PROPOSITION 2 (Kuratowski isometry). *If $P = X$ then $\|\varphi(x) - \varphi(y)\|_\infty = d(x, y)$ for all x, y ; that is, φ is an isometric embedding of (X, d) into $(\ell^\infty, \|\cdot\|_\infty)$.*

Proof. The upper bound is Prop. 1. For the lower bound take the prototype $p = y$: then $|d(x, y) - d(y, y)| = d(x, y)$, so the supremum over prototypes is at least $d(x, y)$. $\square$ $\square$

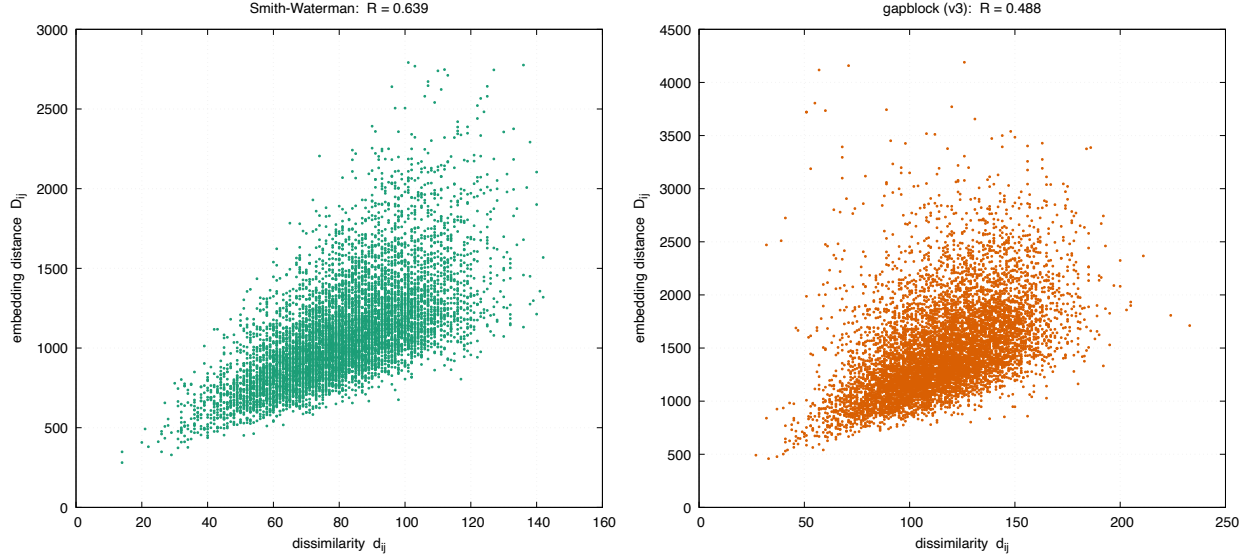


Figure 2: Embedding-space Euclidean distance D_{ij} against pairwise dissimilarity d_{ij} for $n = 2500$ CDR3 β used as their own prototypes (Prop. 3; reproduces suppl. S2). Left: Smith–Waterman d ; right: the gapblock metric shipped in v3. Generated by `mir/bench/theory.py::s2_dissimilarity_distance_correlation`.

The construction of Eq. (1) is thus the Kuratowski/Fréchet embedding, and the supplementary experiments that use the sequences themselves as prototypes (§S1–S2 of [1]) are exactly this isometric regime, read out in ℓ^2 rather than ℓ^∞ . Passing from ℓ^∞ to ℓ^2 costs the $\sqrt{K}$ factor of Eq. (2) but makes the coordinates amenable to PCA and Euclidean clustering.

WHY ℓ^2 DISTANCE TRACKS ALIGNMENT DISTANCE. Fix a prototype-sampling measure μ on X (in TCREMP, $\mu = P_{\text{gen}}$) and draw $p_1, \dots, p_K$ i.i.d. from μ .

PROPOSITION 3 (Energy functional). *Define the μ -energy dissimilarity $\bar{D}_\mu^2(x, y) := \mathbb{E}_{p \sim \mu}[(d(x, p) - d(y, p))^2]$. Then $\frac{1}{K}D(x, y)^2 \rightarrow \bar{D}_\mu^2(x, y)$ almost surely as $K \rightarrow \infty$, and $\bar{D}_\mu^2$ is a μ -dependent (semi)metric with $0 \leq \bar{D}_\mu^2(x, y) \leq d(x, y)^2$.*

Proof. $\frac{1}{K}D^2 = \frac{1}{K} \sum_k (d(x, p_k) - d(y, p_k))^2$ is an empirical mean of i.i.d. terms bounded by $d(x, y)^2$ (Prop. 1); the strong law gives the limit and the bound, and symmetry plus the triangle inequality for $\|\cdot\|_2$ make $\bar{D}_\mu$ a semimetric. □ □

Proposition 3 says D estimates a fixed monotone-in- d functional $\bar{D}_\mu$: two receptors that are close under d have similar distance profiles to the whole of μ and hence small $\bar{D}_\mu$, while dissimilar receptors have systematically different profiles. This is the mechanism behind supplementary Fig. S2 — the empirical Pearson correlation between D_{ij} and d_{ij} (Fig. 2, $R = 0.64$ under Smith–Waterman, 0.49 under the gapblock metric of §T.2) — and behind S2c/d, where D_{ik} concordates with means of d_{ik} over the prototype set: those means are Monte-Carlo estimates of $\bar{D}_\mu$.

HOW MANY PROTOTYPES. The $\sqrt{K}$ gap of Eq. (2) is worst-case; for the *random* landmark map the relative fluctuation of $\frac{1}{K}D^2$ around $\bar{D}_\mu^2$ is $O(K^{-1/2})$ by Prop. 3, so a Johnson–Lindenstrauss-style argument [5] on the bounded coordinates gives the following.

PROPOSITION 4 (Landmark sample complexity). *Let $S \subset X$ with $|S| = n$. For $K = O(\varepsilon^{-2} \log n)$ i.i.d. prototypes, with high probability every pair $x, y \in S$ satisfies $|\frac{1}{K}D(x, y)^2 - \bar{D}_\mu^2(x, y)| \leq \varepsilon d(x, y)^2$ simultaneously.*

Sketch. Each coordinate difference $(d(x, p) - d(y, p))^2 \in [0, d(x, y)^2]$ is bounded; Hoeffding gives a $\exp(-2K\varepsilon^2)$ tail for the deviation of the K -average from $\bar{D}_\mu^2$ per pair, and a union bound over the $\binom{n}{2}$ pairs closes with $K = \Theta(\varepsilon^{-2} \log n)$. $\square$ $\square$

The logarithmic dependence on n is the theoretical counterpart of supplementary Fig. S4, where the UMAP of an antigen-specific set stabilises once $K \gtrsim 100$ prototypes are used: the embedding geometry saturates at a prototype count that grows only logarithmically with the data. Prototype embeddings are the *empirical* / Nyström feature map of the alignment kernel, and Landmark MDS [6, 7] and the dissimilarity representation [8] are the classical instances; the PCA step of §T.4 is Landmark MDS on φ .

T.2. THE ALIGNMENT DISSIMILARITY AS A METRIC

THE GRAM CONSTRUCTION. TCREMP scores two sequences by an alignment similarity s and converts it to a dissimilarity by

$$d(a, b) = s(a, a) + s(b, b) - 2s(a, b). \quad (3)$$

This is exactly the squared distance $\|\psi(a) - \psi(b)\|^2$ induced by a feature map ψ whenever s is an inner product, $s(a, b) = \langle \psi(a), \psi(b) \rangle$. The `v3` implementation (`mir/distances/junction.py`) reads d off directly: `seqtree's` `blosum62()` is already the Gram transform of Eq. (3), and `gapblock.score_matrix` returns d as a genuine metric (zero diagonal, symmetric, non-negative), selecting the best of contiguous gap-block placements at offsets (3, 4, −4, −3).

DEFINITION 1. A symmetric d with zero diagonal is of *negative type* (equivalently, $-d$ is conditionally positive definite) if $\sum_{i,j} c_i c_j d(x_i, x_j) \leq 0$ for all finite $\{x_i\}$ and all $\{c_i\}$ with $\sum_i c_i = 0$.

PROPOSITION 5 (When Eq. (3) is an honest distance). *d of Eq. (3) is of negative type if and only if the centred similarity matrix s is positive semidefinite, in which case $\sqrt{d}$ embeds isometrically into a Hilbert space and $d = \|\psi(\cdot) - \psi(\cdot)\|^2$ for the associated feature map ψ .*

Sketch. This is Schoenberg's theorem [4]: d is of negative type iff e^{-td} is a positive definite kernel for all $t > 0$ iff $d = \|\psi(x) - \psi(y)\|^2$ for a Hilbert map ψ ; expanding the square recovers Eq. (3) with $s(a, b) = \langle \psi(a), \psi(b) \rangle$, so negative type of d is the positive semidefiniteness of the centred s . See [9, 10] for the modern bi-Lipschitz refinements. $\square$ $\square$

ALIGNMENT SCORES NEED CORRECTION. General pairwise-alignment similarities (Smith–Waterman, BLAST) are *not* positive semidefinite, so Eq. (3) can violate $d \geq 0$ or the triangle inequality; canonicalising an alignment score into a metric requires an explicit correction [11]. In v3 this correction is the non-negativity clamp inside `seqtree`’s Gram transform — a projection of the score onto the nearest conditionally-negative-definite cone — which guarantees the metric axioms Prop. 1–2 rely on. The choice of s (BLOSUM62 here; tcrBLOSUM or a VDJ-specific matrix are alternatives) selects *which* metric; all are admissible provided the clamp restores negative type.

GAP-BLOCK VERSUS SMITH–WATERMAN. The v3 junction metric replaces a full Smith–Waterman optimisation by the best of four predetermined contiguous gap-block placements. When the optimal CDR3 alignment inserts a single contiguous gap inside the hypervariable core — the generic case once the conserved Cys/Phe anchors are fixed — the gap-block score equals the Smith–Waterman optimum; the two diverge only for multi-gap alignments of very unequal lengths. Empirically the induced coordinate systems are strongly but not perfectly aligned (Fig. 2: the same D – d correlation is $R = 0.64$ for Smith–Waterman and $R = 0.49$ for `gapblock`), so v3 embeddings are a *new, versioned* coordinate system rather than a bit-for-bit reproduction of the published one.

T.3. IMMUNE RECEPTORS AND THE RECOMBINATION MODEL

THE PROTOTYPE MEASURE IS P_{gen} . The diversity of an immune repertoire is astronomically large, but it is not arbitrary: it is the image of a low-entropy stochastic process, the V(D)J recombination model P_{gen} [2, 12, 3]. TCREMP takes the prototype-sampling measure of §T.1 to be exactly this model, so each coordinate $\varphi(x)_k = d(x, p_k)$, $p_k \sim P_{\text{gen}}$, measures the alignment distance from x to a typical product of recombination.

PROPOSITION 6 (Distance-profile functional). *For $p \sim P_{\text{gen}}$ the empirical distribution of $\{\varphi(x)_k\}_{k=1}^K$ converges weakly to the law of $d(x, p)$. Hence $\varphi(x)$ is a Monte-Carlo sketch of the map $x \mapsto \text{Law}_{p \sim P_{\text{gen}}} d(x, p)$, and two receptors with the same distance-profile-to- P_{gen} are asymptotically embedding-indistinguishable.*

Proof. The coordinates are i.i.d. evaluations of the fixed 1-Lipschitz map $p \mapsto d(x, p)$ under P_{gen} ; the Glivenko–Cantelli theorem gives uniform convergence of their empirical law to that of $d(x, p)$. $\square$ $\square$

This reframes what the embedding “learns”: not a language model of receptor sequences — which the entropy of P_{gen} (~ 40 bits for TRB) makes futile — but each receptor’s geometric relationship to the generative distribution itself. It also predicts that the embedding is insensitive to how the prototypes are obtained, provided they cover the same support.

PROPOSITION 7 (Prototype-source robustness). *Let μ, ν be two prototype measures that are mutually absolutely continuous with bounded density ratio $\rho = d\mu/d\nu$ on the support of the repertoire. Then the energy dissimilarities of Prop. 3 satisfy $\bar{D}_\mu^2(x, y) = \mathbb{E}_\nu[\rho(p)(d(x, p) - d(y, p))^2]$, so the two embeddings induce distances that agree up to the χ^2 discrepancy between μ and ν ; as $\chi^2(\mu \parallel \nu) \rightarrow 0$ the pairwise-distance vectors become perfectly correlated.*

Sketch. Importance sampling rewrites the μ -expectation as a ν -expectation with weight ρ ; a Cauchy–Schwarz bound on Cov under the two measures controls the correlation of the resulting distance vectors by $\chi^2(\mu \parallel \nu) = \mathbb{E}_\nu[\rho^2] - 1$. $\square$ $\square$

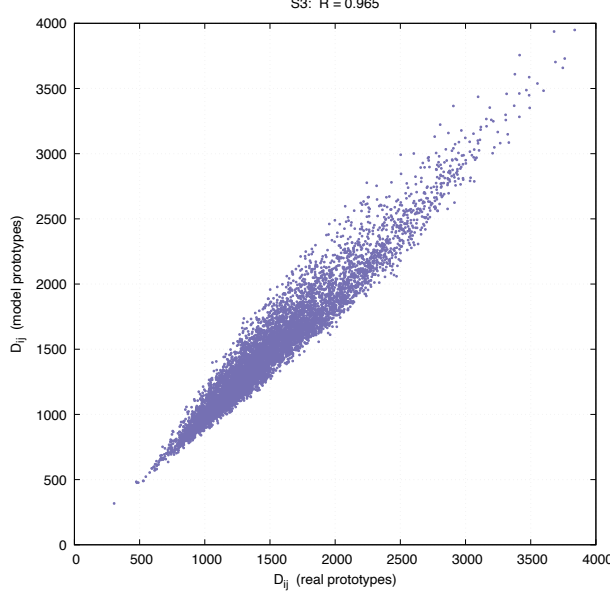


Figure 3: Pairwise embedding distances built from real (Britanova [13]) versus model (OLGA/Murugan [2, 3]) prototypes agree ($R = 0.97$; suppl. S3), confirming Prop. 7. `mir/bench/theory.py::prototype_source_correlation`.

A real healthy-donor repertoire (Britanova *et al.* [13]) and the synthetic recombination model (Murugan *et al.* [2]) are, after thymic selection, close in exactly this sense, which is why supplementary Fig. S3 finds their embedding distances correlated at $R = 0.96$ — reproduced here at $R = 0.97$ (Fig. 3).

DISTRIBUTION LAWS. The random construction also fixes the shape of the two histograms that dominate any embedding analysis.

PROPOSITION 8 (Dissimilarity and distance laws). *Under mild mixing of per-position alignment contributions: (i) the pairwise dissimilarity d_{ij} is asymptotically Gamma-distributed — a sum of many non-negative, bounded per-column penalties, so the central-limit tendency is tempered by positivity and right-skew; (ii) the embedding distance D_{ij} follows a generalized extreme value (Fréchet–Gumbel) law rather than a Gaussian, because $D = \|\varphi(x) - \varphi(y)\|_2$ is dominated by its largest coordinate differences, which are extreme-value statistics over the K prototypes.*

Sketch. (i) d is a sum of per-aligned-column Gram penalties (Eq. (3)), each in a bounded non-negative range; a Gamma law is the maximum-entropy fit to a positive sum with matched mean and variance, and dominates the Gaussian on the skew (Fig. 4, left). (ii) By Prop. 2 the ℓ^∞ part of D equals $d(x, y)$, an extreme over prototypes; the Fisher–Tippett–Gnedenko theorem [14] forces the maximum of K light-tailed coordinate differences into the GEV family, and the same tail governs ℓ^2 once distances concentrate in high dimension [15] (Fig. 4, right). $\square$ $\square$

Both laws are confirmed on v3 (Fig. 4): the Gamma fit to d_{ij} beats the Gaussian on AIC, and the GEV fit to D_{ij} beats the Gaussian decisively (Kolmogorov–Smirnov 0.021 versus 0.078). The

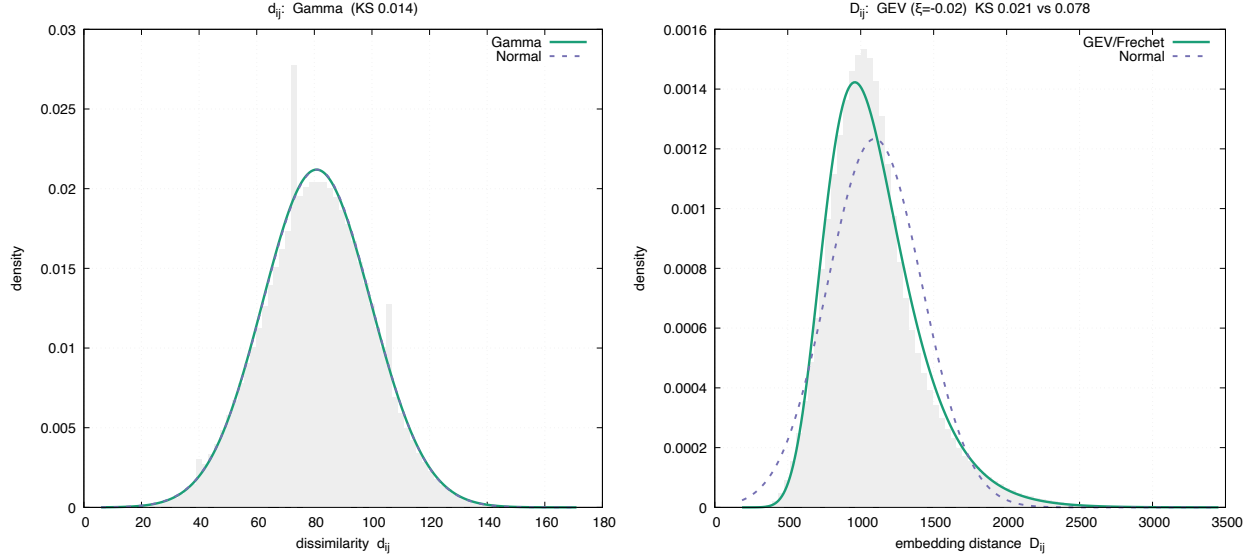


Figure 4: Distribution laws (Prop. 8; suppl. S1). Left: d_{ij} is Gamma, not Gaussian. Right: D_{ij} follows a GEV/Fréchet law (κ s 0.021) that the Gaussian (0.078) misses on the heavy right tail. `mir/bench/theory.py::fit_distributions`.

fitted shape $\xi \approx 0$ places v_3 at the Gumbel–Fréchet boundary, marginally lighter-tailed than the $\xi = +0.11$ of the published Smith–Waterman construction — a metric-dependent detail, not a qualitative difference.

WHY NOT LEARN THE REPRESENTATION? A natural alternative to Eq. (1) is to train a self-supervised sequence model — a masked or autoregressive “receptor language model” — and use its latent as the embedding. The recombination model shows why this cannot improve on the alignment geometry for specificity. Two properties of P_{gen} are decisive: it is *low-complexity* and it is *antigen-blind*.

First, the repertoire process has little learnable structure. P_{gen} factorises as a shallow Bayesian network — germline gene choice, a nearest-neighbour (dinucleotide Markov) insertion chain, and independent trimming [2, 12] — so its *predictive information*, the mutual information between a CDR3 prefix and its remainder (the excess entropy of the process), is bounded by the description length of that network: a few bits, and short-range. Natural language carries large predictive information — syntax, semantics, world knowledge — that deep representations discover and compress; a CDR3 carries almost none beyond its germline templating and local insertion statistics. A likelihood-trained model therefore converges to a sufficient statistic for P_{gen} itself and has nothing further to represent.

Second, and decisively, P_{gen} has no antigen variable: generation precedes and is independent of antigen selection. Let A be the antigen a receptor X recognises, unobserved during self-supervised training.

PROPOSITION 9 (No unsupervised separation of specificity). *Let a representation f be fitted by any objective that is a functional of the unlabelled generation marginal P_{gen} alone. Then the training signal carries zero information about the antigen partition, $I(\text{objective}; A) = 0$; the population objective is invariant under relabelling of antigens, and the alignment of $f(X)$ with A is fixed by the inductive*

bias of f (its architecture, tokenisation, and metric), not by learning.

Sketch. The loss depends on X only through P_{gen} , which is antigen-blind (A enters only at selection, after generation); the data-processing inequality gives $I(f; A) \leq I(\text{training data}; A) = 0$ for any f measurable with respect to the unlabelled sample. Hence the population objective is unchanged by any permutation of antigen labels, so its minimisers cannot prefer an antigen-separating f over an antigen-mixing one on the strength of the data. $\square$ $\square$

Specificity separation in a receptor embedding is thus an *inductive-bias* phenomenon, not a learned one: it is present exactly insofar as the chosen geometry matches the biophysics of recognition. The prototype embedding builds in precisely the right bias — the alignment metric that supplementary Fig. S5 shows tracks structural ($C\alpha$ RMSD) similarity — with no training and no antigen labels. A learned network has only three options: (i) recover P_{gen} , which does not separate antigens (Prop. 9); (ii) be supervised by scarce antigen labels, abandoning the unsupervised premise and the unlimited P_{gen} -sampling that makes the approach attractive; or (iii) regress onto the alignment geometry itself — which is exactly the Part 2 forward codec (`mir/ml/encoder.py`), a network trained to reproduce φ . Option (iii) is not a competitor to the prototype embedding but an accelerator of it: the network learns to *compute* the distance functional quickly, not to discover a better coordinate, and its embeddings are only as good as the φ it approximates. In this precise sense the alignment-to-reference geometry is the canonical unsupervised coordinate for a repertoire whose only structure is an antigen-blind generative model, and prototypes are its minimal, assumption-free realisation.

SPECIFICITY IS HOST-CONDITIONED AND, IN THE LIMIT, COVERAGE-COMPLETE. Proposition 9 concerns unlabelled objectives; one might hope to inject the missing antigen signal from a labelled database. Three facts make even that route converge back to P_{gen} .

(1) *The label is not a function of the receptor.* What one observes as antigen-specific expansion in a donor is not a map $X \mapsto \text{antigen}$ but the host-restricted shadow $A_h(X) = B(X) \cap (\text{HLA}_h \times \text{Exposure}_h)$, where $B(X)$ is the large, cross-reactive set of pMHC that X can engage: which peptides are presented is set by the HLA haplotype, which are encountered by pathogen history. Both are host latent variables, and donors with identical infections hold sparse, *largely disjoint* receptor sets that nonetheless cover the same antigens [16]. Specificity is thus not a stable, sequence-intrinsic function: the same X carries different labels in different hosts, so the map is non-identifiable from sequence across a population.

(2) *Labels cover a vanishing slice.* Databases such as VDJDB [17] annotate a measure-zero corner of the antigen $\times$ receptor space ($\gtrsim 20^9$ peptides per MHC class, thousands of alleles, $> 10^{15}$ possible receptors), so a supervised model saturates at database coverage and collapses on unseen epitopes — it learns the covered epitopes, not specificity.

(3) *The completeness limit is P_{gen} .* Pursuing full coverage does not escape the loop, because the recombination model is itself the immune system’s coverage code: the repertoire is (near-)optimally tuned to the pathogen distribution, exploiting cross-reactivity so that essentially every antigen has cognate precursors [16, 2].

PROPOSITION 10 (Coverage fixed point). *Assume coverage-completeness: there is $p_0 > 0$ such that for almost every antigen a , $\Pr_{x \sim P_{\text{gen}}}[x \text{ binds } a] \geq p_0$. Then the binding relation is a dense, many-*

to-many graph whose antigen-marginal is bounded away from 0 and 1 uniformly; averaging the specificity map over the whole antigen space carries no receptor-discriminative information beyond generation propensity, so as the labelled antigen set expands to the full space the population optimum of any supervised specificity representation converges to the antigen-blind P_{gen} -sufficient statistic of Prop. 9.

Sketch. Coverage-completeness with cross-reactivity makes each receptor bind $\Theta(|\mathcal{A}|)$ antigens and each antigen be bound by a $\geq p_0$ fraction of the repertoire, so the antigen-marginal of the binding indicator is uniformly bounded inside $(0, 1)$; its mutual information with X , integrated over $\mathcal{A}$, is therefore governed by the clone-frequency structure P_{gen} imposes rather than by any low-dimensional antigen axis, and the complete-coverage limit of the supervised objective coincides with the generation-sufficient representation. $\square$ $\square$

The circle closes: an unsupervised model recovers P_{gen} and cannot separate antigens (Prop. 9); a supervised model learns only its labelled slice (fact 2) of a host-conditioned, non-identifiable label (fact 1); and the ideal of learning *all* specificity returns to P_{gen} (Prop. 10), because P_{gen} is the coverage code. Sampling prototypes from P_{gen} is therefore not an arbitrary reference choice — it discretises the very coverage design that antigen breadth is built on, so distance-to-prototypes coordinates a receptor by its position in that coverage geometry, the maximal host-independent, specificity-relevant structure a sequence can carry.

T.4. TCRs, THREE-COMPONENT EMBEDDINGS, AND PAIRED CHAINS

THE PRODUCT METRIC. TCREMP emits three distances per prototype — a V-germline, a J-germline, and a junction distance (or CDR1, CDR2, junction) — interleaved into $\mathbb{R}^{3K}$ (`mir/embedding/tcremp.py`). Writing $\varphi = (\varphi_V, \varphi_J, \varphi_{\text{junc}})$ for the three blocks, the block structure of the Euclidean norm gives the exact decomposition

$$D(x, y)^2 = D_V(x, y)^2 + D_J(x, y)^2 + D_{\text{junc}}(x, y)^2, \quad (4)$$

so the embedding metric is the orthogonal sum of the per-component metrics.

PROPOSITION 11 (Paired-chain concatenation). *For a paired receptor (α, β) embedded by concatenating the per-chain maps, $D_{\text{pair}}(x, y)^2 = D_\alpha(x, y)^2 + D_\beta(x, y)^2$. More generally, concatenating landmark embeddings of independent factors realises the ℓ^2 product of the factor metrics.*

Proof. Concatenation stacks coordinate blocks; the squared ℓ^2 norm of a stacked vector is the sum of the squared block norms, which is Eq. (4) extended across chains. $\square$ $\square$

Paired $\alpha\beta$ TCR embedding (suppl. S4) and, prospectively, the concatenation of epitope- and MHC-groove coordinates into a single TCR–pMHC vector are therefore principled: each modality contributes an orthogonal block and distances add in quadrature.

THE GERMLINE BLOCKS ARE LOW-RANK. The V and J coordinates depend on the sequence only through its germline gene identity, which explains the redundancy that PCA removes.

PROPOSITION 12 (Germline rank bound). *Let G_V be the number of distinct V alleles present in a dataset S . The V-block matrix $\Phi_V \in \mathbb{R}^{|S| \times K}$ with rows $\varphi_V(x)$ has at most G_V distinct rows, hence $\text{rank } \Phi_V \leq G_V$. The same holds for J with G_J , and $G_V, G_J \ll K$.*

Proof. $\varphi_V(x)_k = d_V(v(x), v(p_k))$ depends on x only through its V allele $v(x)$ (the germline distances are a fixed allele $\times$ allele table, `mir/distances/germline.py`); receptors sharing a V allele have identical V-rows, so there are at most G_V distinct rows and the rank is bounded by G_V . $\square$ $\square$

A human TRB repertoire has $G_V \lesssim 60$, $G_J \lesssim 15$, against K in the thousands: two-thirds of the raw coordinates live in a subspace of dimension $\leq G_V + G_J$. The `StandardScaler`→PCA step (`mir/embedding/pca.py`) recovers this compact basis — Landmark MDS on φ — and an ANOVA of the leading components against gene usage recovers the germline axes, while the junction block carries the residual $\Theta(K)$ -dimensional specificity signal.

STRUCTURAL GROUNDING. That the metric is biologically meaningful, not merely combinatorial, is shown by supplementary Fig. S5: embedding distances between TCR:pMHC complexes correlate with the C α RMSD of their crystal structures. Formally this asserts a Lipschitz map from the sequence metric (X, d) into the structural metric, so that specificity groups — receptors binding one epitope — occupy small-diameter regions of embedding space, and clustering recovers them (Table S1; `mir/bench/metrics.py`).

T.5. IGH AND SOMATIC HYPERMUTATION

B-cell receptors differ from TCRs in one decisive respect: after recombination they undergo somatic hypermutation (SHM), a context-dependent point-mutation process [18]. Write M_m for an operator applying m substitutions to a germline-rearranged sequence.

PROPOSITION 13 (Bounded embedding drift). *There is a constant c (the maximal per-substitution Gram penalty) such that $d(x, M_m x) \leq c m$, and therefore $D(\varphi(x), \varphi(M_m x)) \leq \sqrt{K} c m$. Embedding drift under SHM is linear in the mutation load m .*

Proof. Each substitution changes at most one aligned column of Eq. (3), by at most the bounded penalty range c ; m substitutions change d by at most $c m$, and Prop. 1 propagates this to D with the $\sqrt{K}$ factor. $\square$ $\square$

Proposition 13 has two consequences. First, a clonal lineage under affinity maturation traces a path in embedding space whose step size is set by the per-generation mutation load, so the embedding preserves lineage proximity — the substrate for density and trajectory analysis (§T.6). Second, it explains why IGH is the hardest chain for the neural codec of Part 2 (forward cosine 0.79, exact reconstruction 0.16, against $\geq 0.93/0.50$ for TCR): (i) IGH CDR3s are long, so the fixed-length-40 gap-block tokenisation (`mir/ml/tokenize.py`) truncates an appreciable fraction of the variable middle, discarding information; (ii) SHM inflates the junction distance by $\Theta(m)$, so germline-rearrangement

prototypes sampled from P_{gen} systematically under-cover the mutated support (the density ratio ρ of Prop. 7 becomes large and the robustness guarantee degrades); and (iii) the per-receptor entropy is higher. The remedy the theory suggests is to separate the SHM-robust germline distance (V/J, Prop. 12) from the mutation-inflated junction distance and to augment the prototype set with a mutation model, i.e. to sample prototypes from P_{gen} pushed through M_m rather than from P_{gen} alone.

T.6. DENSITY GEOMETRY AND BACKGROUND SUBTRACTION

The embedding turns a repertoire into a point cloud in $\mathbb{R}^{3K}$ (or its PCA projection), and the natural null model is the pushforward of the generation measure.

DEFINITION 2. Let $f_{\text{gen}} = \varphi_{\#} P_{\text{gen}}$ be the density of embedded prototypes and f_{obs} the density of an observed repertoire on embedding space. The *enrichment* of a region is the density ratio $E(z) = f_{\text{obs}}(z)/f_{\text{gen}}(z)$.

PROPOSITION 14 (Continuous limit of neighbour enrichment). *The graph neighbourhood test of TCRNET/ALICE — comparing the observed count of receptors within an alignment radius r of z to its expectation under P_{gen} — converges, as $r \rightarrow 0$, to the density ratio $E(z)$. Convergent selection is precisely the set where $E(z) > 1$.*

Sketch. The observed and expected neighbour counts in a ball of radius r are $f_{\text{obs}}(z)$ and $f_{\text{gen}}(z)$ times the common ball volume $+o(r^d)$; their ratio tends to $E(z)$ as the volume shrinks, which is the standard consistency of a balloon density-ratio estimator. $\square$ $\square$

Proposition 14 recasts graph-based background subtraction as continuous density-ratio estimation on embedding space, which is the foundation for the TCRNET-without-graphs method of Part 2: estimate f_{gen} once from P_{gen} samples (or from the fast forward codec $\varphi \approx$ neural network, `mir/ml/encoder.py`), estimate f_{obs} by kernel density or a normalising flow, and report $E(z)$ directly — no neighbourhood graph, and a smooth enrichment field rather than per-vertex counts.

T.7. PRIVACY: THE EMBEDDING AS A KEYED, LOSSY REPRESENTATION

Immune repertoires are individually identifying — like genomic data, a person’s TCRs are a stable, near-unique fingerprint — so whether an embedding may be shared in place of raw sequences is a practical question with a clean geometric answer. Reproducing a coordinate requires two secrets: the prototype set $P = \{p_k\}$ and the PCA rotation W fitted to the cohort (together with the centring μ and scaling). Write the released representation as $Z = (\varphi(X) - \mu) W \in \mathbb{R}^{n \times m}$ with $m \ll 3K$.

THREE BARRIERS TO RECONSTRUCTION. (i) *Lossy projection.* PCA to m components discards a $(3K - m)$ -dimensional subspace, so $\varphi \mapsto Z$ is many-to-one and non-invertible on the discarded directions *even given* W — an information-theoretic barrier no decoder can cross. (ii) *Unknown anchors.* Recovering x from $\varphi(x) = (d(x, p_k))_k$ is reconstruction from distances to *unknown* reference points in a discrete sequence space, a hard combinatorial inverse problem without P . (iii) *Data-dependent mixing.* W is estimated from the cohort, not a public transform; without it the released

numbers are linear combinations of unknown prototype distances. Releasing Z alone is therefore a **keyed, lossy pseudonymisation**: the pair (P, W) is a secret key, and without it reconstruction reduces to a hard inverse problem stacked on an irreversible compression.

WHAT IT DOES NOT PROTECT. The embedding is built to preserve pairwise geometry (Prop. 1–3), and that is precisely what a linkage adversary exploits. A key-holder — or anyone able to embed a reference sequence under the same (P, W) — can (a) test membership by embedding a candidate and matching it against Z (re-identification / membership inference), (b) link the same patient across releases through preserved distances, and (c) invert approximately with a trained decoder: the Part 2 inverse codec (`mir/ml/decoder.py`) demonstrates that φ is approximately invertible *given the key*. Distance preservation is thus double-edged — the source of the embedding’s analytic value and of its linkage risk. Keyed pseudonymisation is not anonymisation.

A DIFFERENTIAL-PRIVACY MECHANISM THE GEOMETRY HANDS US. The Lipschitz property of Prop. 1 turns the embedding into a clean substrate for a formal guarantee. Replacing one receptor moves its coordinate by $\|\varphi(x) - \varphi(x')\|_2 \leq \sqrt{K} d(x, x') \leq \sqrt{K} d_{\max}$, so $X \mapsto \varphi(X)$ has bounded ℓ^2 sensitivity.

PROPOSITION 15 (Private release of the embedding). *Releasing $\tilde{Z} = (\varphi(X) - \mu)W + N$ with N Gaussian of scale $\sigma \propto \sqrt{K} d_{\max} \|W\|_2 / \varepsilon$ satisfies record-level (ε, δ) -differential privacy, and by Prop. 1 the induced distortion of any pairwise embedding distance D_{ij} is $O(\sigma\sqrt{m})$ — a privacy/utility trade-off governed entirely by the sensitivity constant $\sqrt{K} d_{\max}$.*

Sketch. The bounded ℓ^2 sensitivity of φ composed with the linear map W gives Gaussian-mechanism sensitivity $\sqrt{K} d_{\max} \|W\|_2$; the Gaussian mechanism then yields (ε, δ) -DP for σ as stated, and the triangle inequality bounds the perturbation of each D_{ij} by the noise magnitude $O(\sigma\sqrt{m})$. $\square \quad \square$

RECOMMENDATION. For patient data, treat (P, W) as a managed secret, release only the truncated projection Z (never φ or the invertible residual), and — when a provable guarantee is required rather than obscurity — add the calibrated noise of Prop. 15, whose cost is a quantifiable, Lipschitz-bounded distortion of the very distances the analysis depends on.

T.8. INSIGHTS, REPRODUCTION, AND OPEN PROBLEMS

A UNIFYING VIEW. The results above place TCREMP in a well-understood corner of metric geometry. The map φ is the empirical (Nyström) feature map of the alignment kernel; its ℓ^∞ read-out is the Kuratowski isometry, its ℓ^2 read-out is Landmark MDS, and its PCA is the classical scaling of the landmark columns. Three observations may be of independent interest beyond immunology: (a) taking the landmark measure to be a *generative model* of the data (P_{gen}) rather than a random subsample makes the embedding a sketch of each point’s relation to the data-generating process, and makes prototype choice provably robust (Prop. 7); (b) high-dimensional prototype-embedding distances obey a generalized extreme value law (Prop. 8), a clean consequence of distance concentration that is easy to mistake for Gaussian; and (c) the germline rank bound (Prop. 12) shows how categorical side-information injects exactly-low-rank structure into an otherwise full-rank distance embedding.

REPRODUCTION. Every quantitative statement is regenerated by `mir/bench/theory.py` and the data/figure scripts in this directory (`gen_theory_data.py`, the `*.gp` gnuplot scripts, `diagram_embedding.dot`); Table 1 collects the numbers.

Table 1: Theory predictions and their v3 reproduction (`mir/bench/theory.py`; $n = 2500$ CDR $_3\beta$). Paper values from [1], suppl. S1–S3.

Claim	Paper	mirpy (Smith–Waterman)	mirpy (gapblock v3)
S2 / Prop. 3: $\text{corr}(D_{ij}, d_{ij})$	0.56	0.64	0.49
S1 / Prop. 8: law of d_{ij}	Gamma $>$ Normal	Gamma $>$ Normal	Gamma $>$ Normal
S1 / Prop. 8: law of D_{ij}	GEV, $\xi = +0.11$	GEV $\gg$ Normal, $\xi \approx 0$	GEV $\gg$ Normal, $\xi \approx 0$
S3 / Prop. 7: real vs. model prototypes	0.96	—	0.97

OPEN PROBLEMS. Four questions are left open. (1) Sharp distortion constants for the lower bound complementary to Prop. 1 when prototypes are P_{gen} -sampled, i.e. a Bourgain-type [19] guarantee tuned to the recombination measure rather than to the worst case. (2) The exact extreme-value index ξ of Prop. 8 as a function of the substitution matrix and prototype count. (3) A mutation-corrected metric for IGH that makes Prop. 7 hold under SHM (§T.5). (4) Consistency and rates for the density-ratio estimator of Prop. 14 on the PCA-projected embedding, which would put continuous background subtraction on the same footing as the graph tests it replaces.

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